

Task	Prompt	Answer
strongly connected number	The task is to determine the number of strongly connected components in a directed graph. A strongly connected component is a maximal subgraph where every node is reachable from every other node. Here is a directed graph containing nodes from 1 to 6. The edges are: (2, 5), (5, 1), (3, 4), (6, 2). Question: How many strongly connected components are there in the graph? Your answer should be an integer.	6
topological sort	The task is to determine the topological sort of a directed acyclic graph (DAG). Here is a directed graph containing nodes from 1 to 6. The edges are: (1, 6), (1, 5), (1, 4), (1, 3), (1, 2). Question: What is the topological sort of the directed acyclic graph (DAG)? You need to format your answer as a list of nodes, e.g., [node-1, node-2, ..., node-n].	[1, 6, 5, 4, 3, 2]
traveling salesman problem	The task is to determine the minimal cost of the Traveling Salesman Problem (TSP). The Traveling Salesman Problem asks for the shortest possible route that visits each vertex exactly once and returns to the starting vertex. The input graph is guaranteed to be a complete graph. Here is an undirected graph containing nodes from 1 to 8. The edges are: (1, 2, 9), (1, 3, 3), (1, 4, 6), (1, 5, 8), (1, 6, 7), (1, 7, 4), (1, 8, 9), (2, 3, 10), (2, 4, 11), (2, 5, 5), (2, 6, 11), (2, 7, 1), (2, 8, 9), (3, 4, 11), (3, 5, 1), (3, 6, 9), (3, 7, 2), (3, 8, 9), (4, 5, 8), (4, 6, 3), (4, 7, 4), (4, 8, 8), (5, 6, 3), (5, 7, 3), (5, 8, 10), (6, 7, 8), (6, 8, 1), (7, 8, 10). (u ,v, w) denotes the edge from node *u* to node *v* has a weight of *w*. Question: What is the minimal cost of the Traveling Salesman Problem on the graph? You need to format your answer as a float number.	27.0
triangles	The task is to find the number of triangles that include a specific node as one vertex. A triangle is a set of three nodes that are all connected to each other. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 8. The edges are: (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7), (1, 8), (2, 3), (2, 4), (2, 5), (2, 6), (2, 7), (2, 8), (3, 4), (3, 5), (3, 6), (3, 7), (3, 8), (4, 5), (4, 6), (4, 7), (4, 8), (5, 6), (5, 7), (5, 8), (6, 7), (6, 8), (7, 8). Question: How many triangles include node 1 in the graph? Your answer should be an integer.	21